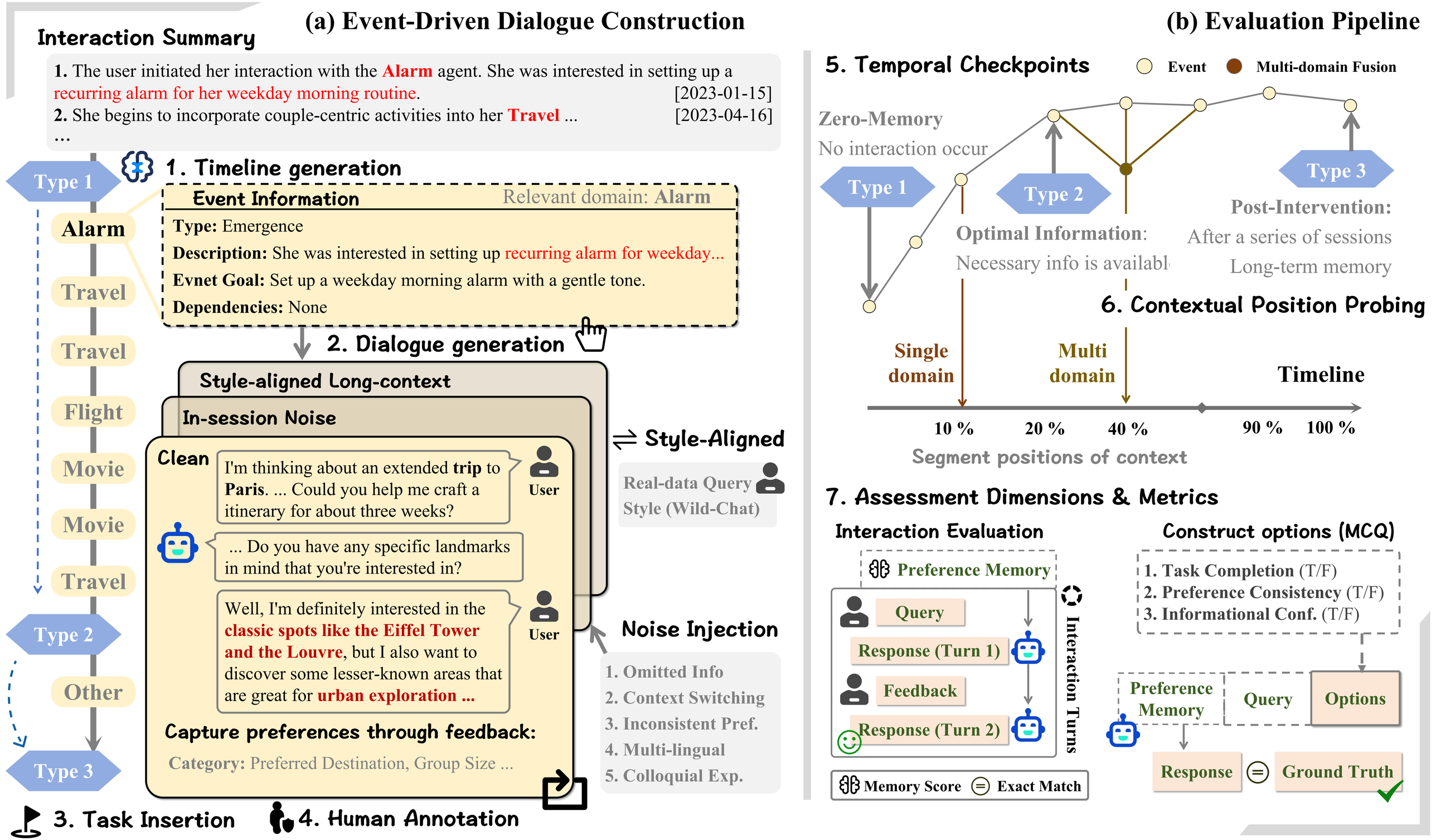
(a) Event-Driven Dialogue Construction
(b) Evaluation Pipeline
Interaction Summary
1. The user initiated her interaction with the Alarm agent. She was interested in setting up a recurring alarm for her weekday morning routine. [2023-01-15]
2. She begins to incorporate couple-centric activities into her Travel ... [2023-04-16]
...
Type 1
Alarm
Travel
Travel
Flight
Movie
Movie
Travel
Type 2
Other
Type 3
1. Timeline generation
Event Information
Relevant domain: Alarm
Type: Emergence
Description: She was interested in setting up recurring alarm for weekday...
Evnet Goal: Set up a weekday morning alarm with a gentle tone.
Dependencies: None
2. Dialogue generation
Style-aligned Long-context
In-session Noise
Clean
I'm thinking about an extended trip to Paris. ... Could you help me craft a itinerary for about three weeks?
User
... Do you have any specific landmarks in mind that you're interested in?
Well, I'm definitely interested in the classic spots like the Eiffel Tower and the Louvre, but I also want to discover some lesser-known areas that are great for urban exploration ...
User
Capture preferences through feedback:
Category: Preferred Destination, Group Size ...
Style-Aligned
Real-data Query Style (Wild-Chat)
Noise Injection
1. Omitted Info
2. Context Switching
3. Inconsistent Pref.
4. Multi-lingual
5. Colloquial Exp.
3. Task Insertion
4. Human Annotation
5. Temporal Checkpoints
Event
Multi-domain Fusion
Zero-Memory
No interaction occur
Type 1
Type 2
Type 3
Optimal Information:
Necessary info is availabl
Post-Intervention:
After a series of sessions
Long-term memory
6. Contextual Position Probing
Single domain
Multi domain
Timeline
10 %
20 %
40 %
90 %
100 %
Segment positions of context
7. Assessment Dimensions & Metrics
Interaction Evaluation
Preference Memory
Query
Response (Turn 1)
Feedback
Response (Turn 2)
Interaction Turns
Memory Score
Exact Match
Construct options (MCQ)
1. Task Completion (T/F)
2. Preference Consistency (T/F)
3. Informational Conf. (T/F)
Preference Memory
Query
Options
Response
Ground Truth